Map Properties Projection and Transformation

Definition of Map

The early human beings used to transfer their feelings to other persons and community through articulacy (by speaking), literacy (by writing), numeracy (by counting) and graphicacy (by picture).

The present- day map is the product of the skill of graphicacy.

‘a map is worth thousands of words’.

A map consists……

Certain Cartographic elements, which are:

- **conventional signs**
- **drawn to a scale, and**
- **certain projection system**

so that each and every point on it corresponds to the actual terrestrial or celestial position.

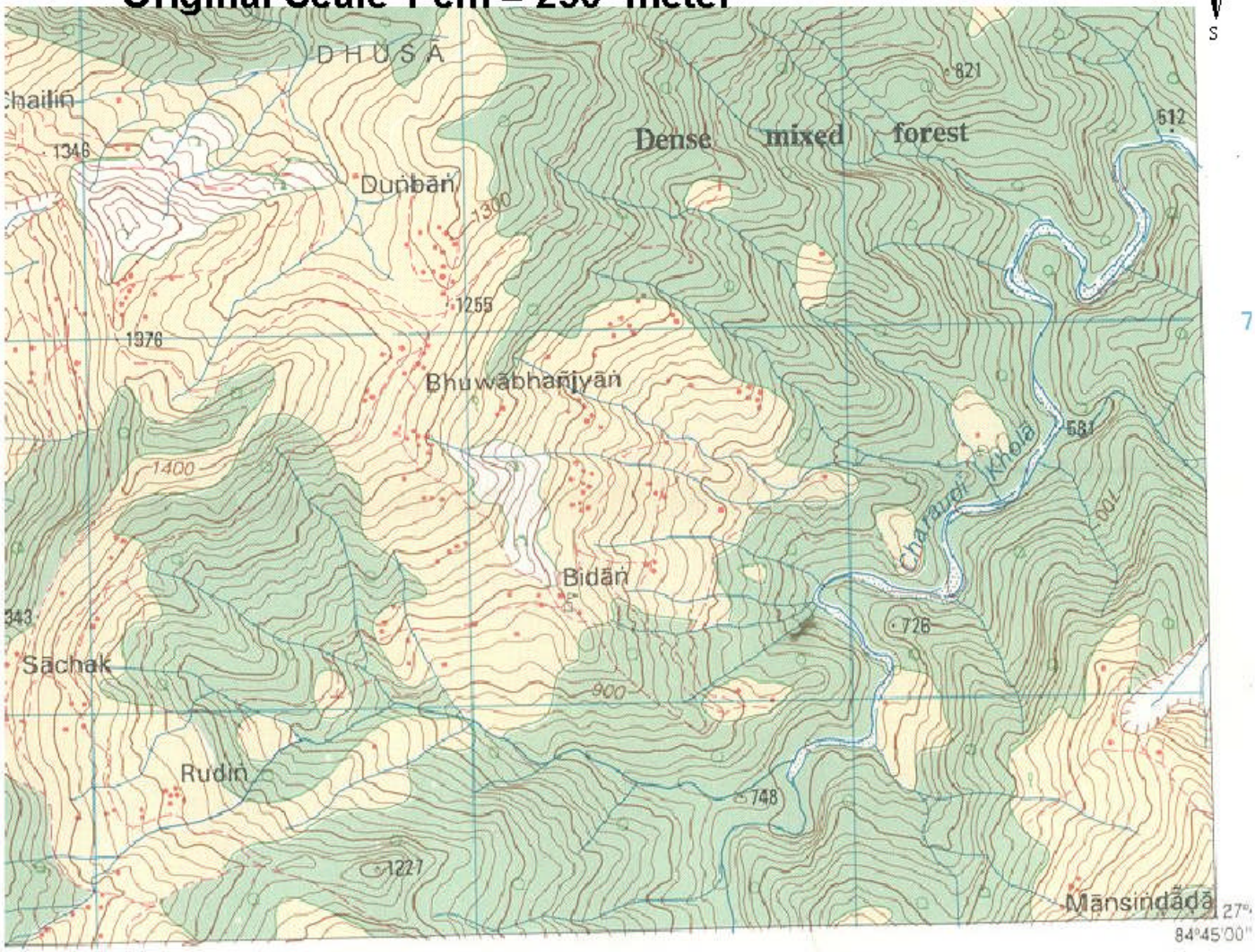
Types of Map

The Types of Map can be grouped by its:

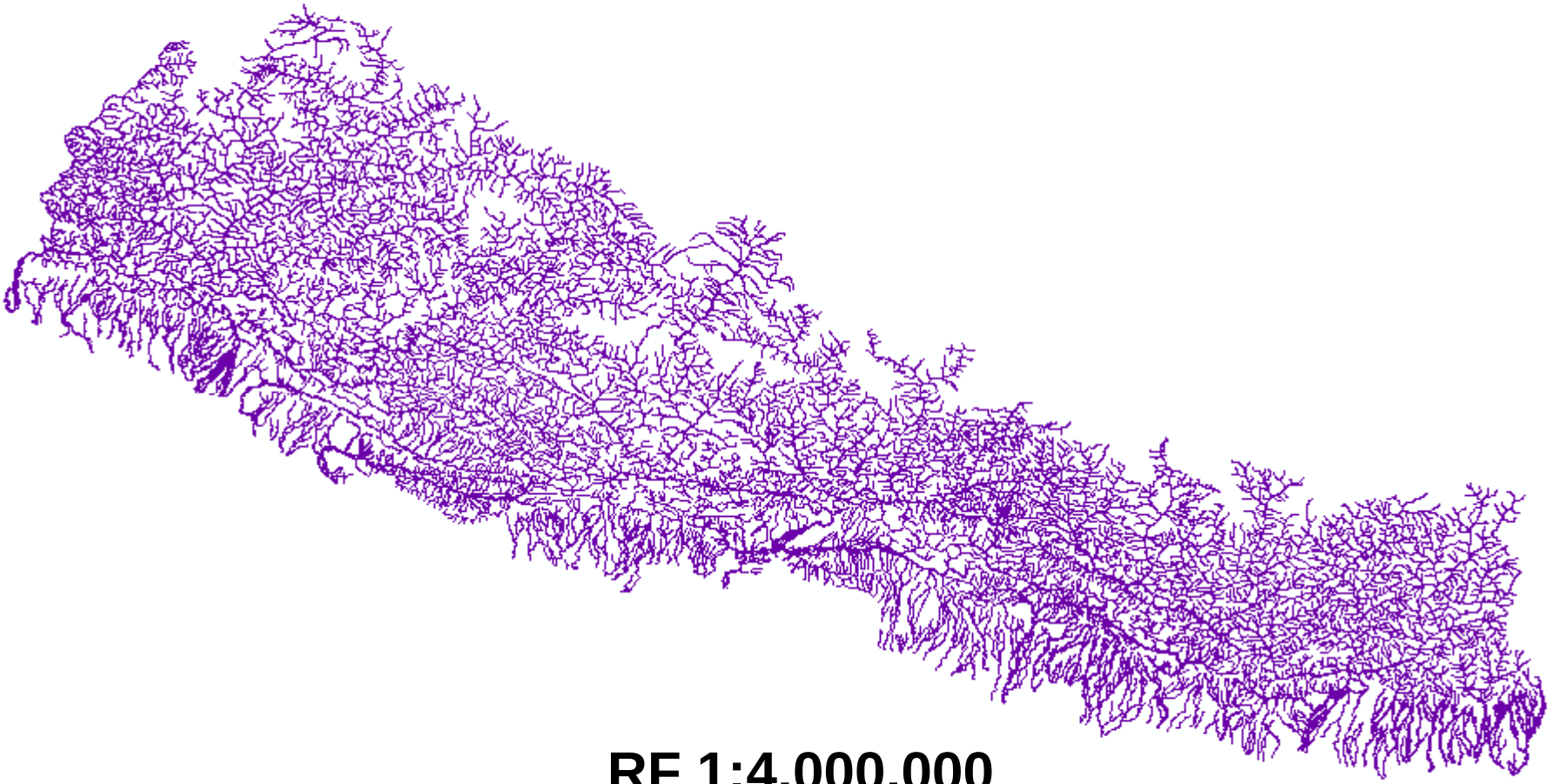
- **scale of representation and**
- **the information/contents which it has carries.**

Large Scale General map

Topographic Map 1994, Sheet No.2784 03D
Original Scale 1 cm = 250 meter



Small Scale Thematic map



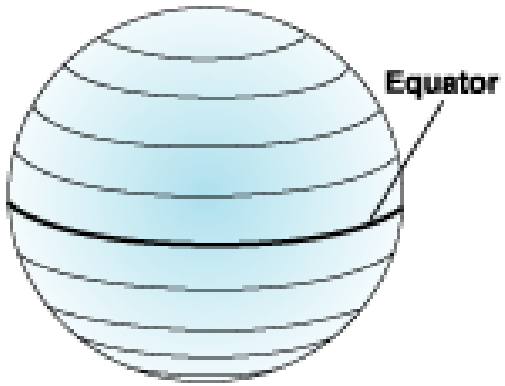
RF 1:4,000,000

Coordinates

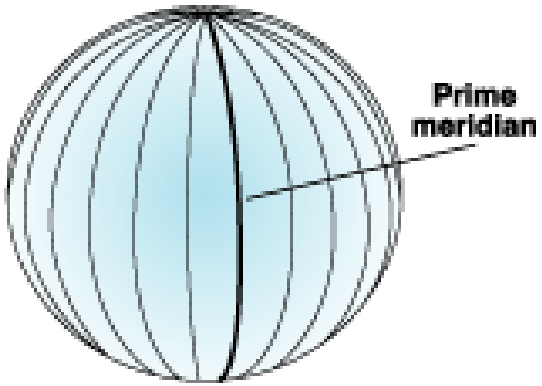
- means a structured set of measurements related to a specific spatial reference system; usually applied to points of distance measurements (X, Y) on independent axes of a planar reference system or to angular measurement such as latitude and longitude pairs on a spherical reference model.

The spherical system: an illustration

This illustration shows the parallels and meridians that form a graticule.



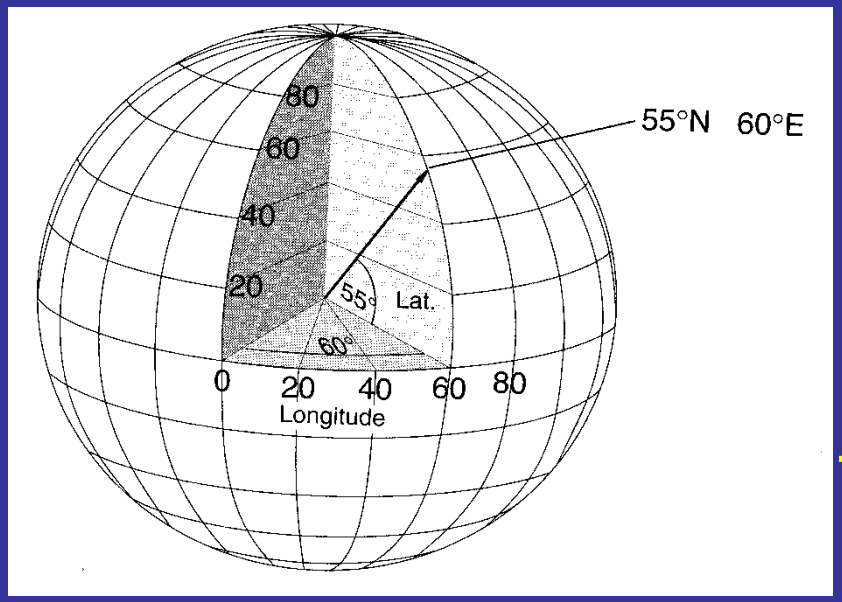
Parallels
(Lines of latitude)

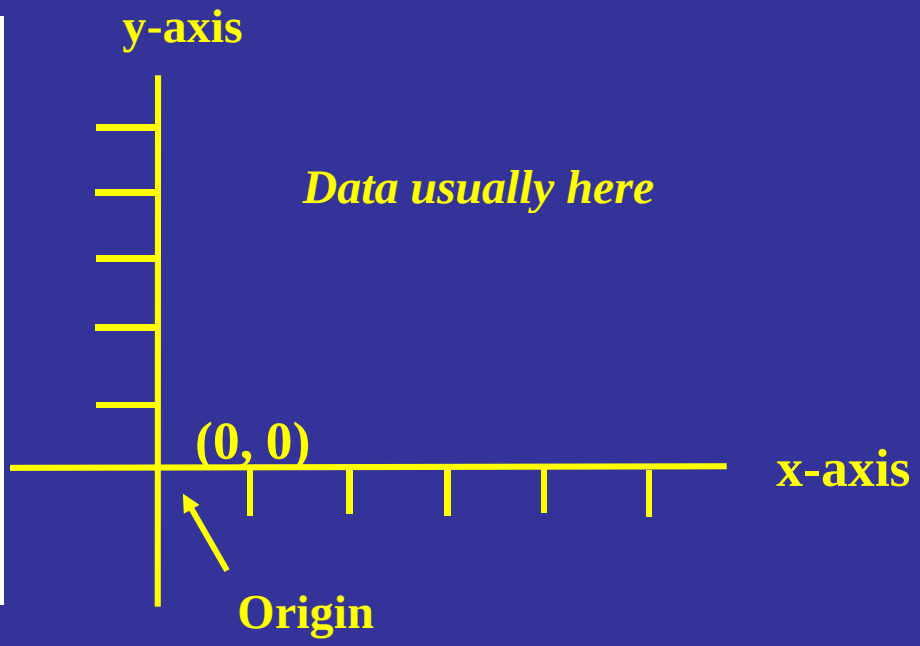


Meridians
(Lines of longitude)

Graticular
Network

Coordinate System

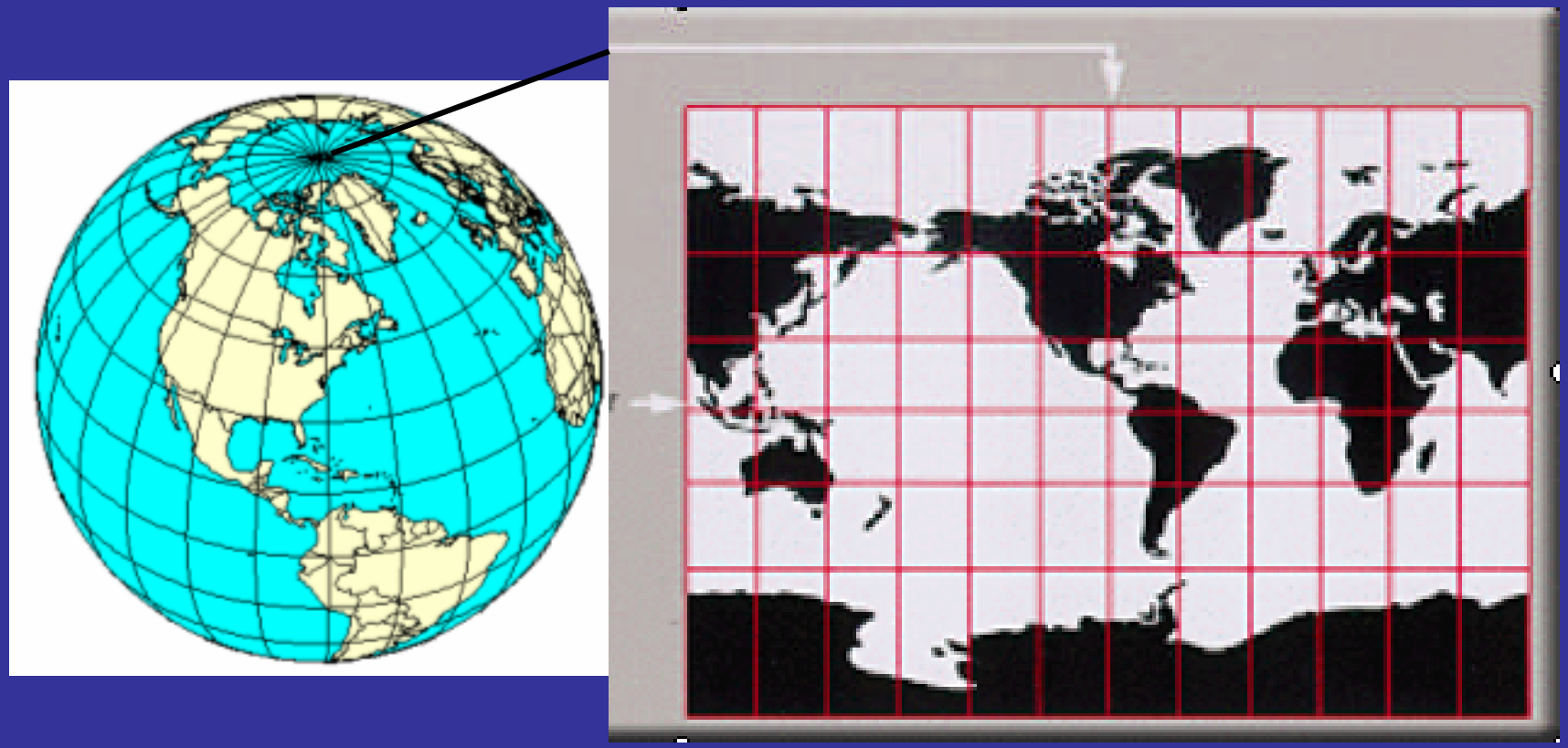




Spherical coordinate system

Cartesian coordinate system

What is map projection?

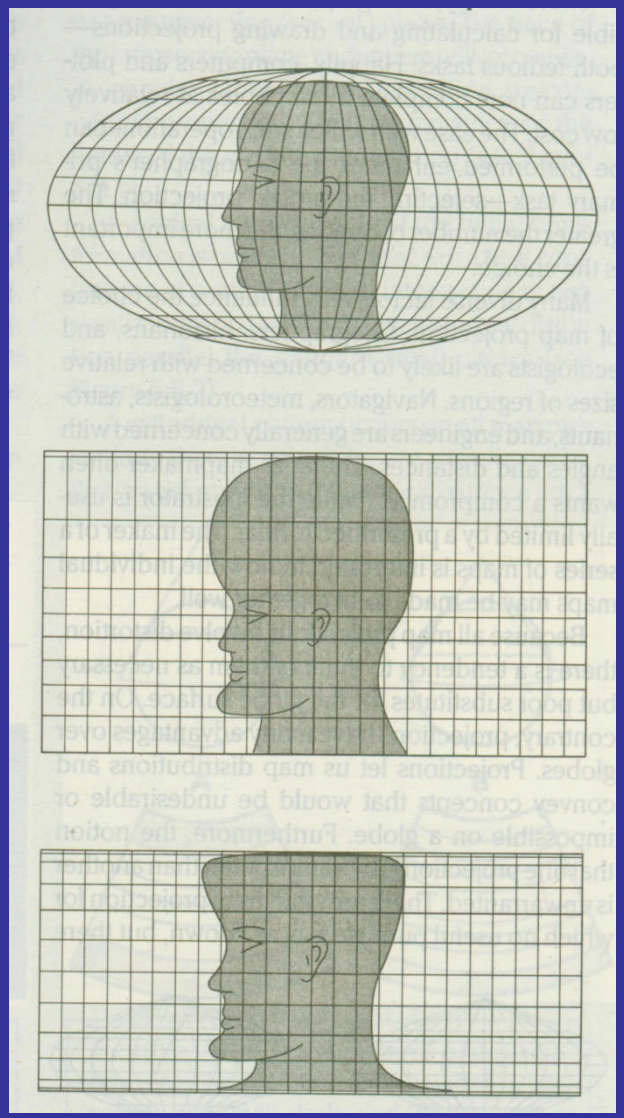


distortion of the shape can be seen in the projected figures

Mollweide

Mercator's

Cylindrical equal area



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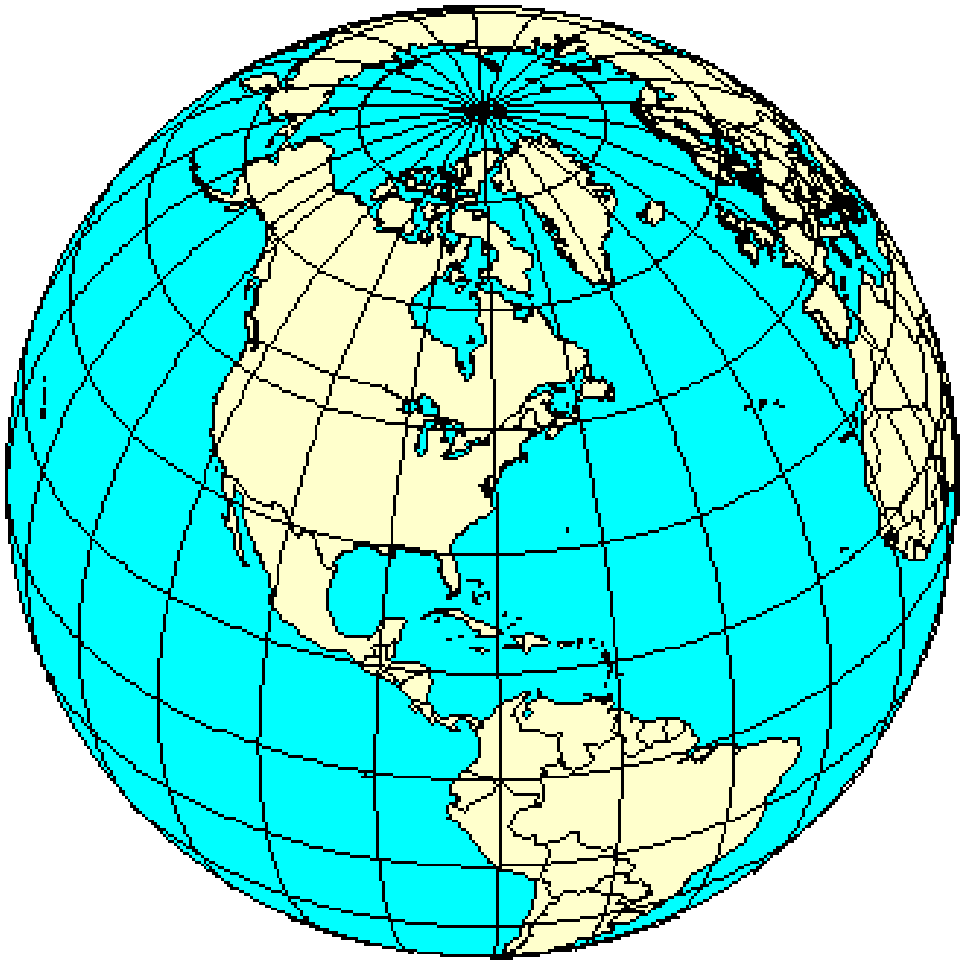
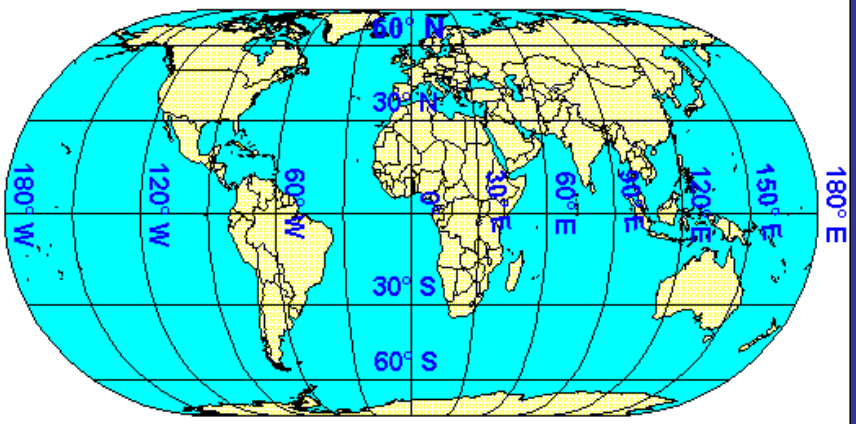
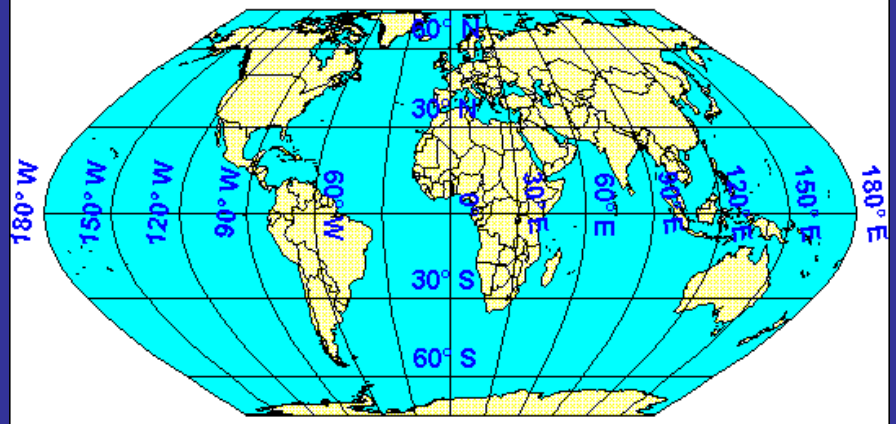


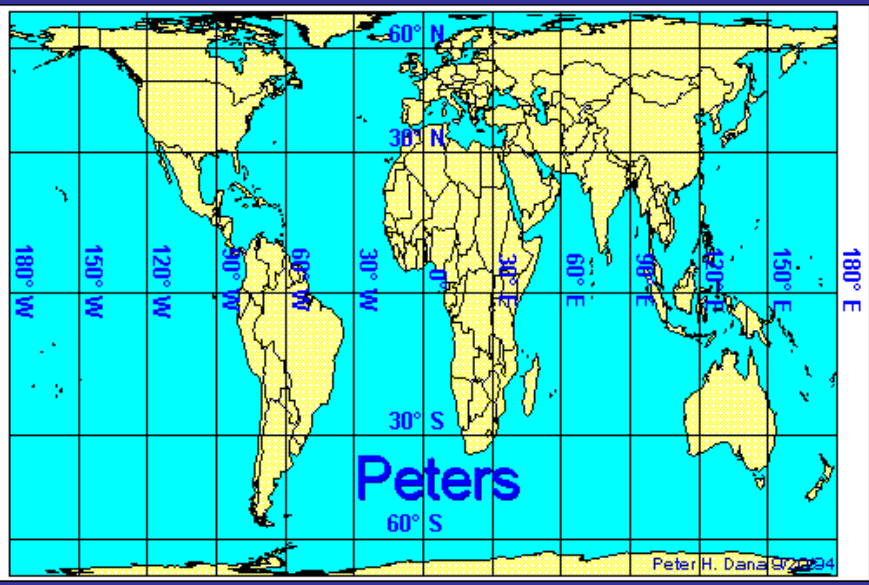
Orthographic Projection
Centered on Washington, DC

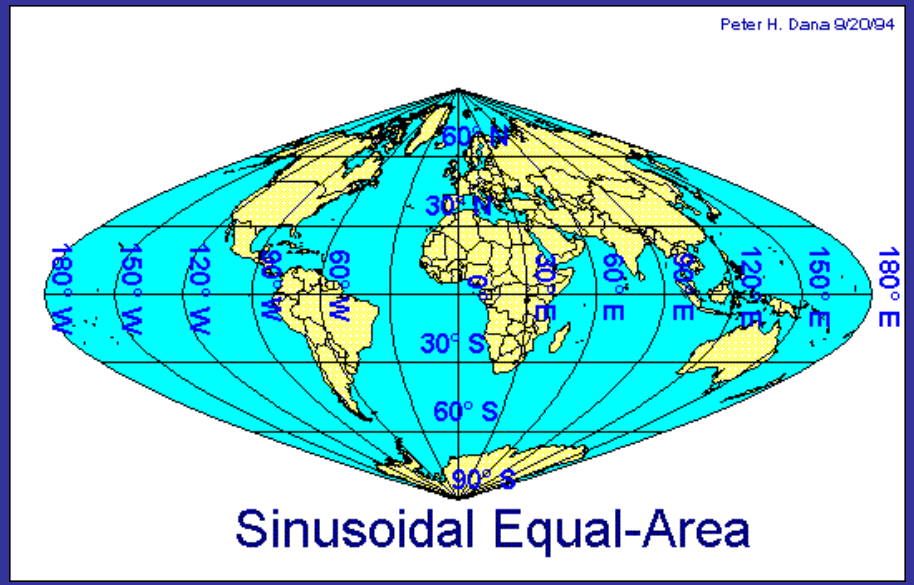


Eckert IV Equal Area

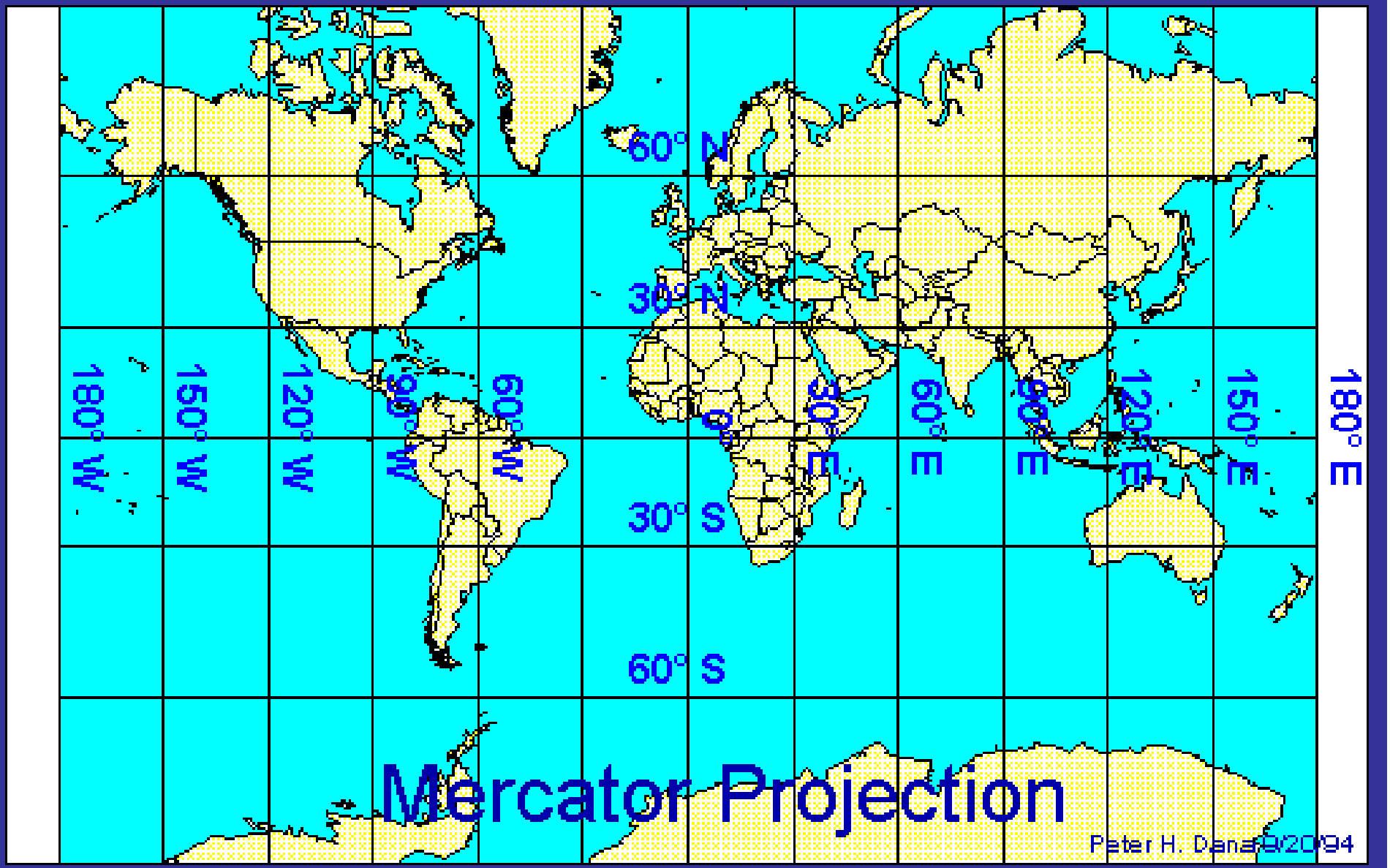


Eckert VI Equal Area





Sinusoidal Equal-Area


Mercator Projection
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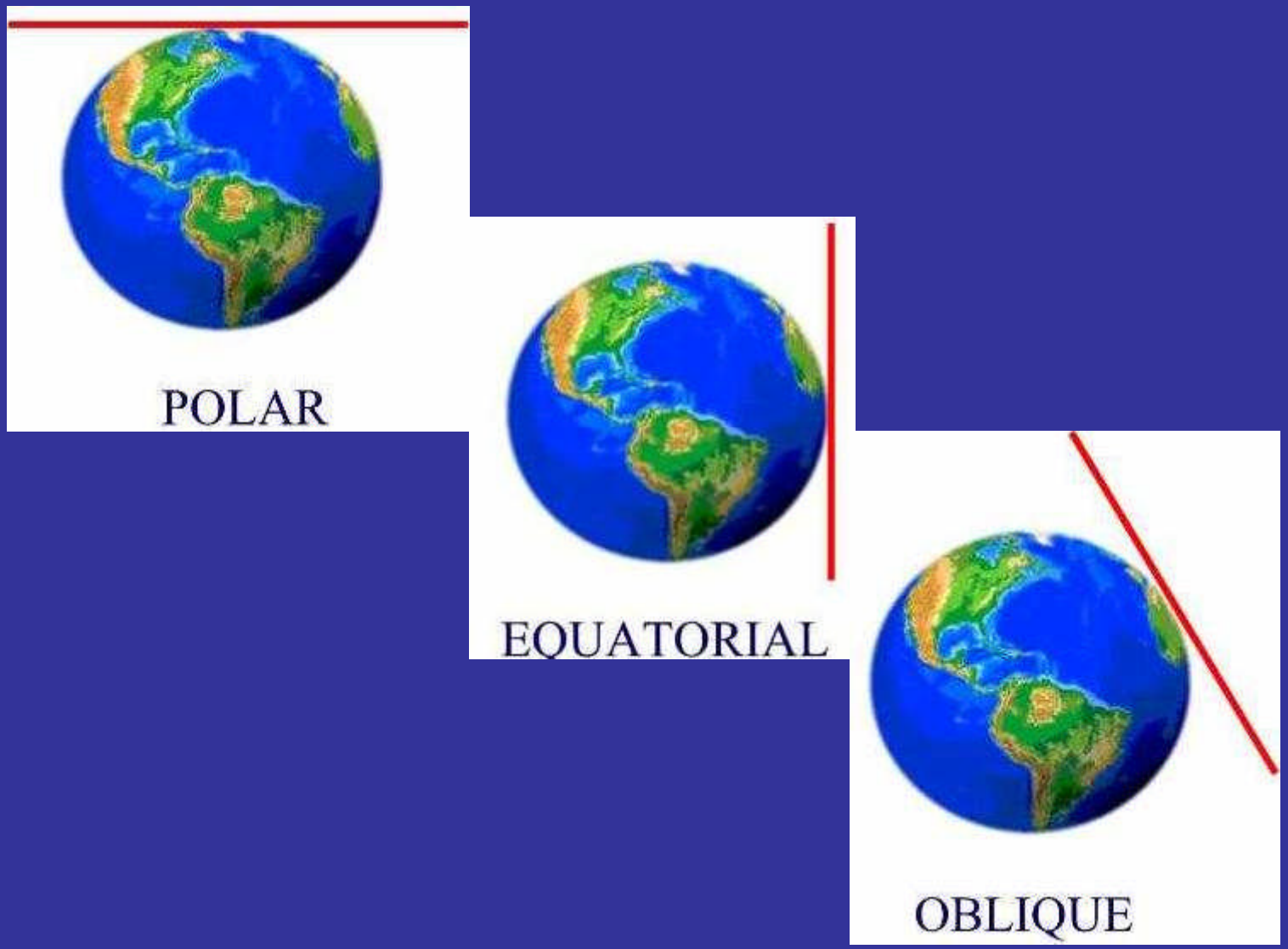
60° N
30° N
0°
30° S
60° S
180° W
150° W
120° W
90° W
60° W
30° E
60° E
90° E
120° E
150° E
180° E

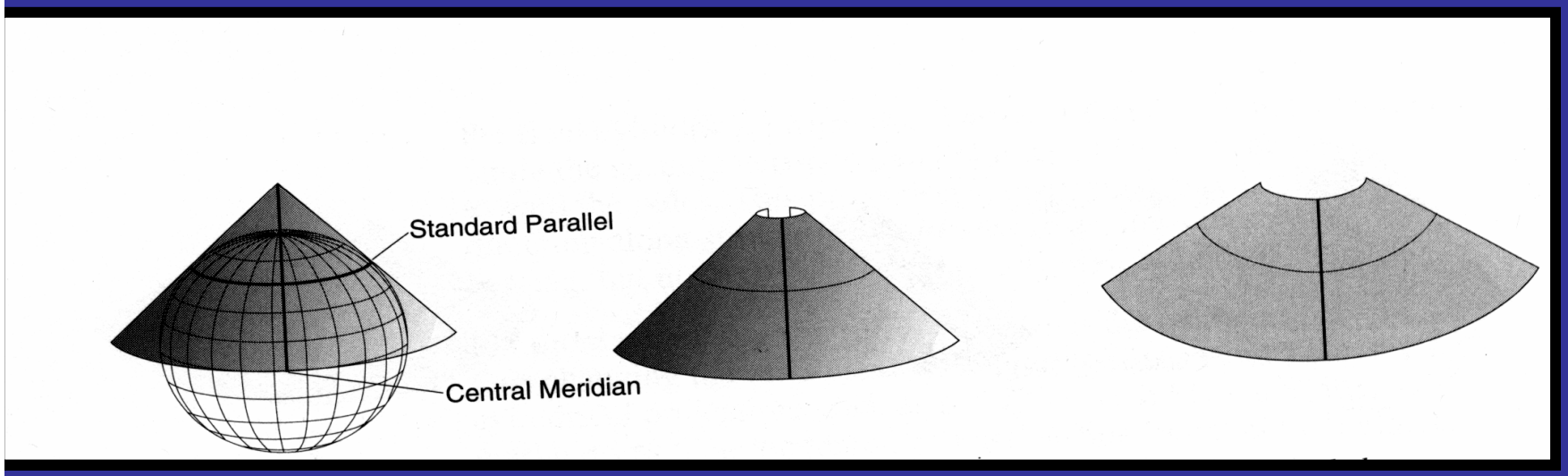
Projections

- Planar or Azimuthal projection
- Conical projection
- Cylindrical Projection

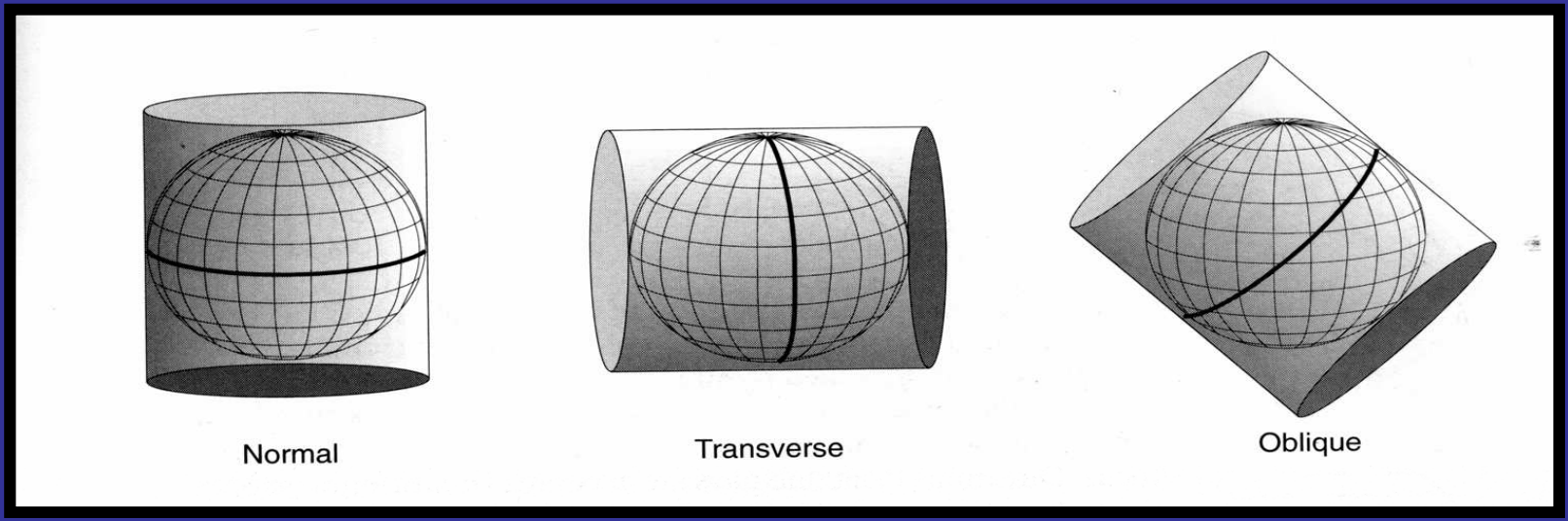
Planar or Azimuthal Projection

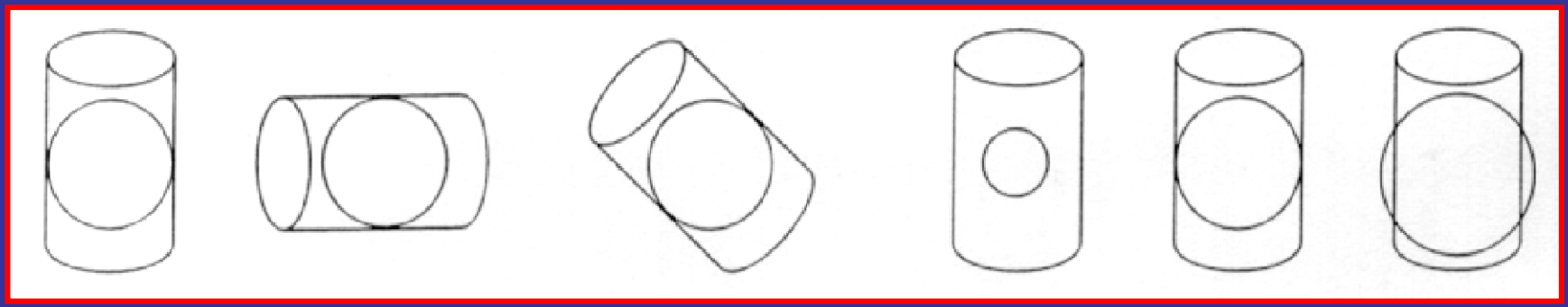


Conic Projection



Cylindrical Projection



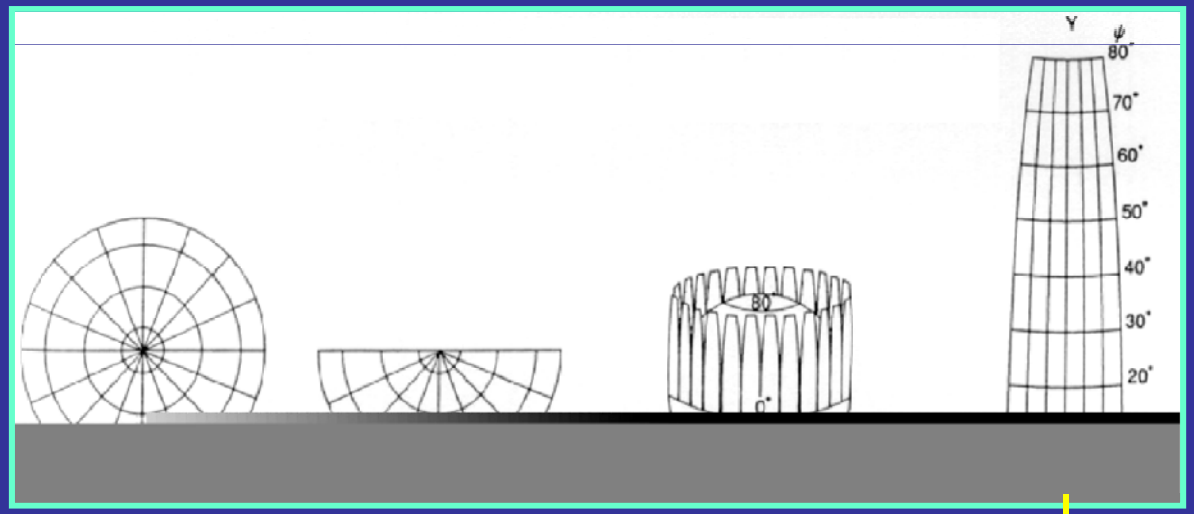


(a) normal (b) transverse (c) oblique (a) inside (b) tangent (c) secant

Cylindrical Projection with the Aspect

Cylindrical Projection with the Cylinder Size

Polar Stereo Projection



Lambert's Conformal Conical Projection

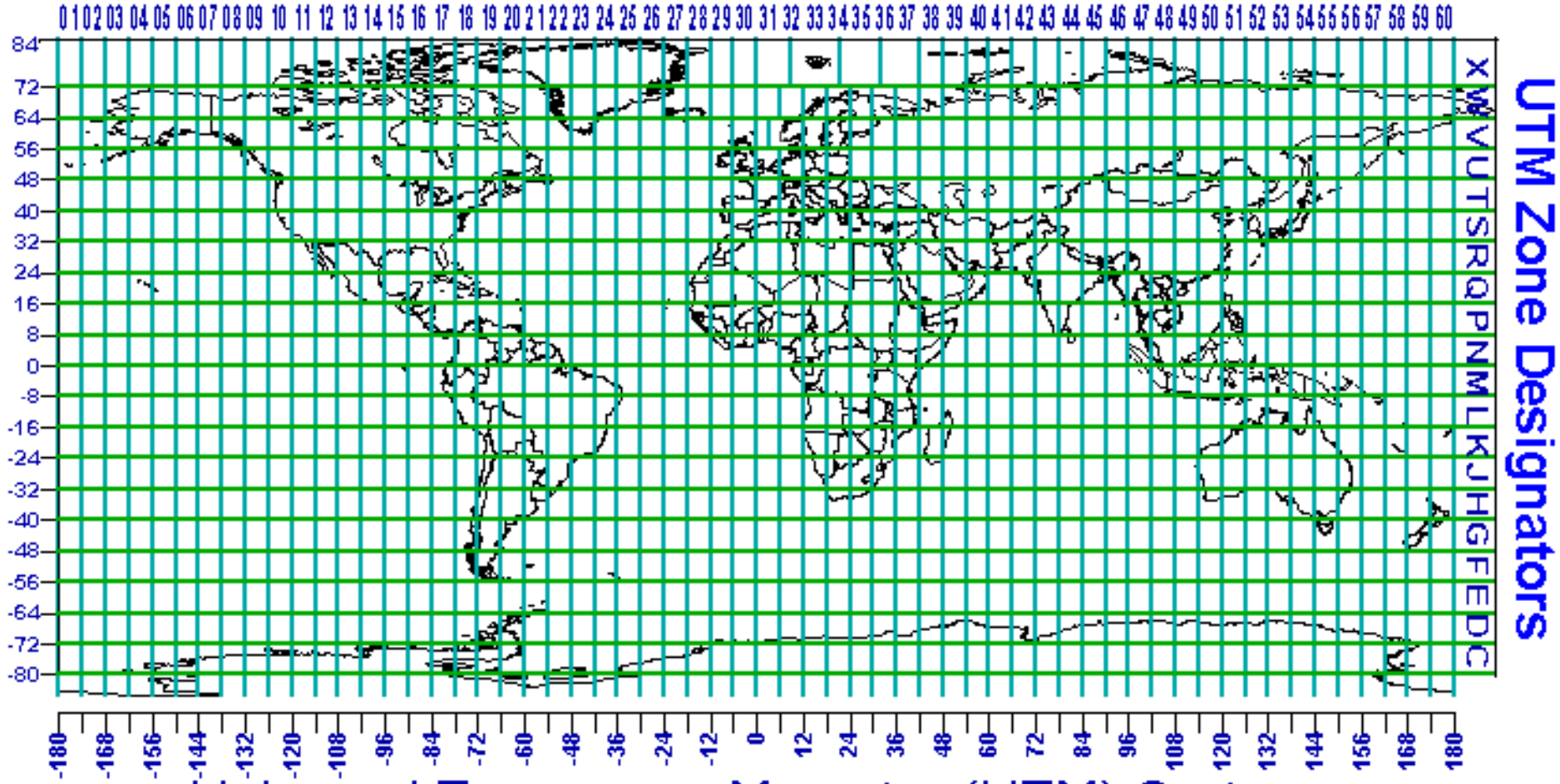
(a) UTM zones in the northern hemisphere

-3^0 0 $+3^0$

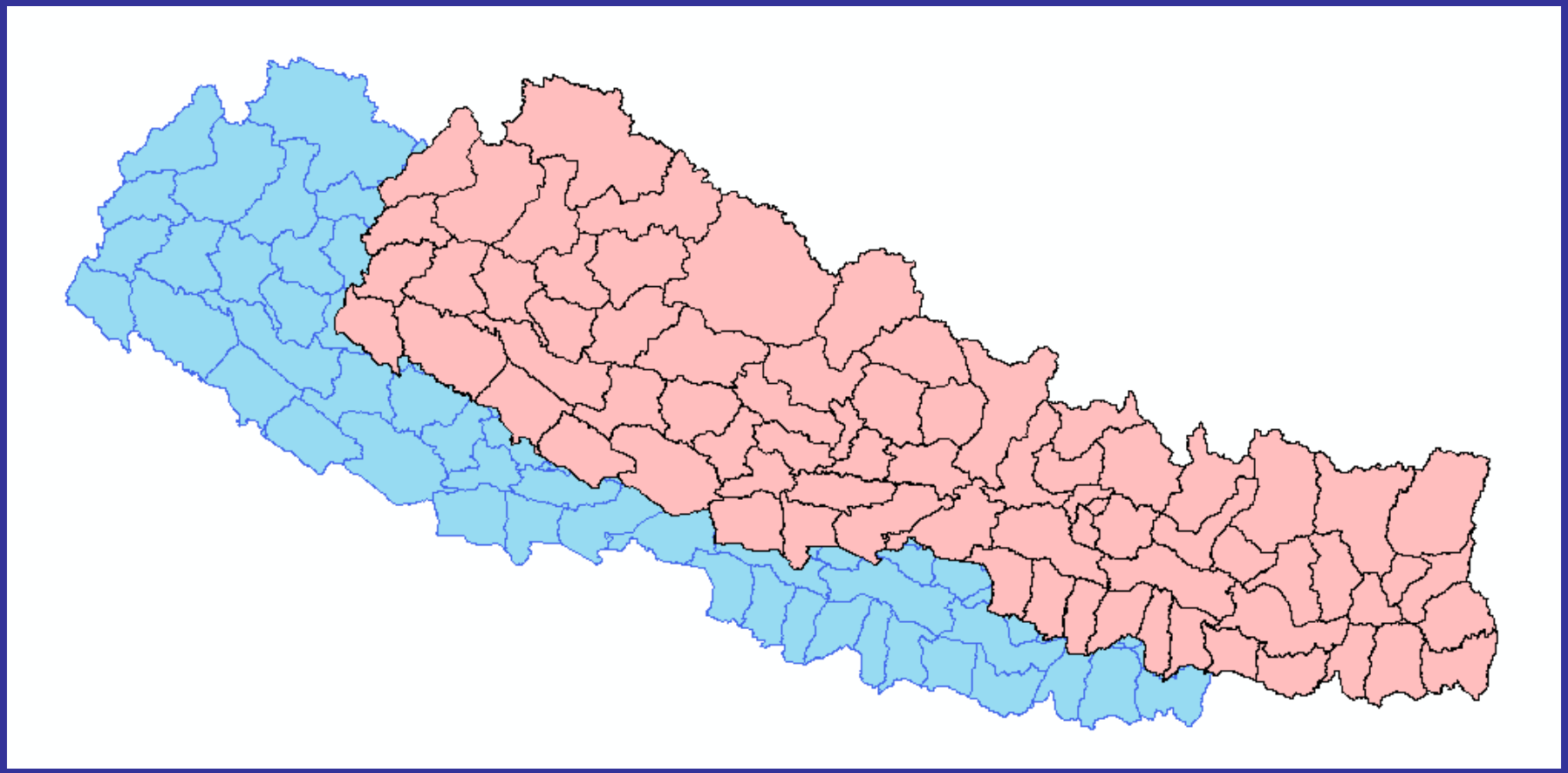
UTM

(b) UTM Coordinate System (Origin:X=5 x10^5m,Y=0)

UTM Zone Numbers



UTM Zone Designators

Universal Transverse Mercator (UTM) System

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